

Team Worksheet — Lab 3

DMAIC at Depth — Recap, $Y = f(X)$ and Baseline Sigma for Your Project

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-03-baseline-sigma-calculator.xlsx`.

3. What we produced

Deliverable: A DMAIC tollgate map, a $Y = f(X)$ model and a baseline DPMO / sigma level for your capstone.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Recap each DMAIC phase in full: its purpose, its core tools, its tollgate deliverable and the most common way teams fail in it.

Step 2. For each phase, write the one question that phase answers — Define: what problem? Measure: how big? Analyze: why? Improve: what fix? Control: how do we hold it?

Step 3. Write your capstone project Y — the single output metric your customer actually feels. It must be measurable and it must already have data.

Step 4. Brainstorm at least twelve candidate Xs that could drive that Y. Classify each as controllable, noise or SOP-governed.

Step 5. Write the relationship formally as $Y = f(X_1, X_2, \dots, X_n)$, and mark which Xs you can experiment on later in the DOE labs.

Step 6. Collect or estimate your baseline: units processed, defects observed and opportunities per unit over a defined recent period.

Step 7. Calculate DPU, DPO and DPMO. Show the formulas: $DPU = \text{defects}/\text{units}$; $DPO = \text{defects}/(\text{units} \times \text{opportunities})$; $DPMO = DPO \times 1,000,000$.

Step 8. Convert DPMO to a baseline sigma level using the conversion table, and state whether the 1.5-sigma long-term shift is included.

Step 9. Record this baseline in your capstone pack — every improvement claim on Day 5 is measured against this number.

7. Self-check

You can explain all five DMAIC tollgates without notes, your Y is a measurable customer-facing output with at least twelve classified Xs, and your baseline DPMO and sigma level are calculated from real counts.

Complete? ■ Yes ■ Not yet

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